Taewan Kim

HCI Research Scientist, TwelveLabs

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Introduction

HCI researcher specializing in human-AI interaction and AI-driven service design, with industry experience at Twelvelabs, Samsung Electronics, and NAVER AI Lab. I earned my Ph.D. from KAIST, where I investigated how foundation models such as large language models and multimodal AI can be applied to real-world services in healthcare, mental wellbeing, and content creation. My work bridges user-centered research, system prototyping, and deployed AI applications, with publications in top venues such as CHI, CSCW, and DIS.

Education

2021 – 2024 **KAIST** – Daejeon, Korea
Ph.D. in Industrial Design
Advisor: Hwajung Hong
Dissertation: *Designing for Mental Wellbeing: Exploring the Role of AI for Reflective Practice*

2018 – 2021 **Seoul National University** – Seoul, Korea
Ph.D. Candidate in Cognitive Science
Completed all coursework; transferred to KAIST following advisor.

2016 – 2018 **UNIST** – Ulsan, Korea
M.S. in Creative Design Engineering

2011 – 2016 **Handong University** – Pohang, Korea
B.S. in Product Design

Industry Experience

Jun. 2025 – Current **TwelveLabs** – San Francisco/Seoul
HCI Research Scientist
- Conducting research on human-AI interaction for multimodal video understanding, focusing on AI-driven content analysis and creation.

May 2024 – Jun. 2025 **Samsung Electronics** – Seoul
Senior Designer
- Explored how generative AI can augment early-stage UX design through synthetic persona-based validation and user studies.

Jan. 2023 – Jul. 2023 **NAVER** – Seongnam-si
Research Intern
- Led research on large language models in mental health contexts, designing and evaluating multimodal AI interfaces through user-centered methods.

Journal Articles / Conference Papers (Peer-Reviewed)

2026 **Enhancing Patient Participation in Emergency Department through Patient-Friendly Clinical Notes Generated by Large Language Models**
Sung-In Kim, Joonyoung Park, Taewan Kim, Wooseok Seo, Tae Rim Kim, Won Chul Cha, Hwajung Hong
Scientific Reports

2025 **Exploring Design Spaces to Facilitate Household Collaboration for Cohabiting Couples**
Gahyeon Bae, Seo Kyoung Park, Taewan Kim, Hwajung Hong
*Proceedings of the CHI Conference on Human Factors in Computing Systems (**CHI**).*

2024 **MindfulDiary: Harnessing Large Language Model to Support Psychiatric Patients' Journaling**
Taewan Kim, Seolyeong Bae, Hyun AH Kim, Su-woo Lee, Hwajung Hong, Chanmo Yang, Young-Ho Kim
*Proceedings of the CHI Conference on Human Factors in Computing Systems (**CHI**).*
Acceptance rate: 26.4%

2024 **DiaryMate: Understanding User Perceptions and Experience in Human-AI Collaboration for Personal Journaling**
Taewan Kim, Donghoon Shin, Young-Ho Kim, Hwajung Hong
*Proceedings of the CHI Conference on Human Factors in Computing Systems (**CHI**).*
Acceptance rate: 26.4%

2024 **Co-Creating Question-and-Answer Style Articles with Large Language Models for Research Promotion**
Hyunseung Lim, Ji Yong Cho, Taewan Kim, Jeongeon Park, Hyungyu Shin, Seulgi Choi, Sunghyun Park, Kyungjae Lee, Juho Kim, Moontae Lee, Hwajung Hong
*Proceedings of the Designing Interactive Systems Conference (**DIS**)*

2024 **Promoting Self-Efficacy of Individuals with Autism in Practicing Social Skills in the Workplace Using Virtual Reality and Physiological Sensors: A Feasibility Study**
Sung-In Kim, So-youn Jang, Taewan Kim, Bogoan Kim, Dayoung Jeong, Taehyung Noh, Mingon Jeong, Kaely Hall, Meelim Kim, Hee Jeong Yoo, Kyungsik Han, Hwajung Hong, Jennifer G. Kim
JMIR Formative Research

2022 **Prediction for Retrospection: Integrating Algorithmic Stress Prediction into Personal Informatics Systems for College Students' Mental Health**
Taewan Kim, Haesoo Kim, Hayeon Lee, Hwarang Goh, Shakhboz Abdigapporov, Mingon Jeong, Hyunsung Cho, Kyungsik Han, Youngtae Noh, Sung-Ju Lee and Hwajung Hong
*Proceedings of the CHI Conference on Human Factors in Computing Systems (**CHI**).*
Acceptance rate: 24.7%

2022 **Sad or just jealous? Using Experience Sampling to Understand and Detect Negative Affective Experiences on Instagram**
Mintra Ruensuk, Taewan Kim, Hwajung Hong, and Ian Oakley
*Proceedings of the CHI Conference on Human Factors in Computing Systems (**CHI**).*
Acceptance rate: 24.7%

2022 **The Workplace Playbook VR: Exploring the Design Space of Virtual Reality to Foster Understanding and Support of Autistic People in the Workplace**
Jennifer G. Kim, Taewan Kim, Sungin Kim, Soyeon Jang, Stephanie Lee, Heejung Yoo, Kyungsik Han, and Hwajung Hong
*ACM Hum.-Comput. Interact. 6, CSCW2 (**CSCW**)*

2022 **VISTA: Understanding Characteristics of Autistic People through a VR-based Interactive Social Skills Training System**
Bogoan Kim, Dayoung Jeong, Mingon Jeong, Taehyung Noh, Sung-In Kim, Taewan Kim, So-youn Jang, Hee Jeong Yoo, Jennifer G Kim, Hwajung Hong, and Kyungsik Han
*Proceedings of the 28th ACM Symposium on Virtual Reality Software and Technology. (**VRST**)*
Acceptance Rate: 24.4%

2021 **Understanding University Students' Experiences, Perceptions, and Attitudes Toward Peers Displaying Mental Health-related Problems on Social Network Sites: Online Survey and Interview Study**
Taewan Kim and Hwajung Hong
JMIR Mental Health

2020 **In Helping a Vulnerable Bot, You Help Yourself: Designing a Social Bot as a Care-Receiver to Promote Mental Health and Reduce Stigma**
Taewan Kim, Mintra Ruensuk, and Hwajung Hong
*Proceedings of the CHI Conference on Human Factors in Computing Systems (**CHI**).*
Acceptance Rate: 24.3%

2018 **Design Constraints and Their Influence Upon Design Outcome**
Taewan Kim, James Andrew Self, and Hwajung Hong
Archives of Design Research 2018.

Extended Abstracts, Poster (Lightly Peer-Reviewed)

2024 **Evaluating and Auditing LLM-Driven Chatbots for Psychiatric Patients in Clinical Mental Health Settings**
Taewan Kim, Seolyeong Bae, Hyun AH Kim, Su-woo Lee, Hwajung Hong, Chanmo Yang, Young-Ho Kim
CHI 2024 Workshop on Human-centered Evaluation and Auditing of Language Models

2024 **Towards an Evaluation of LLM-Generated Inspiration by Developing and Validating Inspiration Scale**
Hyungyu Shin, Seulgi choi, Ji Yong Cho, Sahar Admoni, Hyunseung Lim, Taewan Kim, Hwajung Hong, Moontae Lee, Juho Kim
CHI 2024 Workshop on Human-centered Evaluation and Auditing of Language Models

2023 **DiaryMate: Exploring the Roles of Large Language Models in Facilitating AI-mediated Journaling**
Taewan Kim, Donghoon Shin, Young-Ho Kim, and Hwajung Hong
CHI 2023 Workshop on Intelligent and Interactive Writing Assistants(In2Writing)

2023 **Designing a Personalized Stress Management System for Call Center Workers**
Kwangyoung Lee, Hyunseung Lim, Sooyeon Ahn, Taewan Kim, and Hwajung Hong
IEEE 2023 International Conference on Big Data and Smart Computing (BIGCOMP)

2021 **Leveraging Challenges of an Algorithm-Based Symptom Checker on User Trust Through Explainable AI**
Youjin Hwang, Taewan Kim, Junhan Kim, Joonhwan Lee, and Hwajung Hong
CHI 2021 Workshop on Realizing AI in Healthcare: Challenges Appearing in the Wild

2021 **An Exploratory Study on the Algorithm User Experience of a Symptom Checker Application for Self-Diagnosis**
Taewan Kim, Youjin Hwang, Junhan Kim, Joonhwan Lee, and Hwajung Hong
The Proceedings of HCI KOREA 2021

2018 **Studying Students Experiencing Mental Health Problems**
Taewan Kim, and Hwajung Hong
CSCW 2018 Workshop on Conducting Research with Stigmatized Populations

2017 **Calm Station: An Interactive Perpetual Desk Object that Reduces Digital Distractions**
Taewan Kim, Young-Woo Park, and Hwajung Hong
Proceedings of the 2017 ACM Conference Companion Publication on Designing Interactive Systems (DIS).

2015 **FamCom: A Communication Service Enhancing Conversation Quality Between Elders Residing in Care Hospital and Their Family Member**
Mingu Kanng, Taewan Kim, Youngjae Kim, and Junghwan Ahn
Proceedings of the 33rd Annual ACM Conference Extended Abstracts on Human Factors in Computing Systems (CHI).

Invited Talks

Apr. 2024 Harnessing Large Language Model to Support Psychiatric Patients' Journaling
NAVER Tech Meetup

Apr. 2024 The Potential for Generative AI in Psychiatry
2024 Korean Neuropsychiatric Association Spring Conference
(대한신경정신의학회 춘계학술대회)

Feb. 2024 Harnessing Large Language Model for Enhancing Communication Between Patients and Psychiatrists
2024 Korean Neuropsychiatric Association Jeonbuk Branch
(대한신경정신의학회 전북지부 학회)

Jul. 2023 Design for Mental Wellbeing: Exploring the Role of Artificial Intelligence for Self-Reflection
HAI LAB at Seoul National University of Science and Technology

Jul. 2023 Writing a Diary with AI: A New Way to Understand Myself
NAVER UX Designer Workshop

Jan. 2023 Exploring the Role of Large Language Models in Retrospective Writing
NAVER Tech Talk

Teaching experience

Spring 2022 **Introduction to UI/UX, Prototyping, and Usability Evaluation**
KAIST Course for Hana Bank employees
(KAIST 하나은행 비학위과정 UI/UX 디자인 개론), **Lecturer**

Fall 2022 Smart Health: Data-Driven Service Design for Health and Wellbeing
KAIST CS492(36.492), Teaching assistant (taught in English)

Fall 2019 Creative Research Design
Seoul National University (SNU) 2114.409, Teaching assistant

Academic Service

Journal and Conference Reviewer
- ACM Conference on Human Factors in Computing Systems (CHI)
- ACM Conference on Computer Supported Cooperative Work and Social Computing (CSCW)
- Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)
- Conference on Language Modeling (COLM)
- Archives of Design Research (ADR)
- International Association of Societies of Design Research (IASDR)
- ACM Conference on Designing Interactive Systems (DIS)
- ACM Conference on Tangible and Embedded Interaction (TEI)

Associate Chair
- ACM Conference on DIS 2025 Provocations and Work-in-Progress (PWiP)

References

Hwajung Hong
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Assistant Professor, School of Interactive Computing, Georgia Institute of Technology
jennifer.kim@cc.gatech.edu

Kyungsik Han
Associate Professor, Department of Data Science, Hanyang University
kyungsikhan@hanyang.ac.kr